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Core Java Interview Questions by Shambhu Sir

@javac_java

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Basic Concept:

1. What is programming language?
2. What are the different types of programming language?
3. Explain high level programming language?
4. High level language is of how many types?
5. **Explain difference between procedure oriented and object-oriented programming?**
6. Explain difference between compiler and interpreter.
7. **What are the four pillars of object-oriented programming?**
8. **Define all four pillars such as Encapsulation, Inheritance, Polymorphism and Abstraction.**
9. How java is a platform independent language?
10. What is Byte Code/magical code in java?
11. **Explain the working of JVM?**
12. What are the types of Error?
13. Explain compile time error and Run time Error.
14. Explain what is exception?
15. What is exception handling?

Structure of Java Program:

16. **What will happen if a java program doesn't have main method?**
17. What are keywords? Mention few of them.
18. How many keywords are used in main method of java?
19. What are identifiers? Tell me all its rules.
20. What is difference between keywords and identifiers?
21. **Why String[] args is used in main method of java? What is its use?**
22. **Explain command line argument feature.**
23. **What will happen if a program has multiple methods?**
24. **What will happen if there are multiple main method inside a program with different arguments?**

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➢ Data Types:

25. Explain data types. Why it is used?
26. How many data types are in java? Explain them.
27. **Mention at least 5 differences between primitive and non-primitive data.**
28. **Represent byte a=100; in memory representation of 2's compliment.**
29. **Represent byte a=80; in memory representation of 2's compliment.**
30. **Represent byte a=-120; in memory representation of 2's compliment.**
31. **Represent byte a=-28; in memory representation of 2's compliment.**
32. **Design a non-primitive data for Employ with some attributes and show how to use it.**
33. **Design a non-primitive data for Product with some attributes and show how to use it.**
34. **Design a non-primitive data for Vehicle with some attributes and show how to use it.**
35. **Design a non-primitive data for Fruit with some attributes and show how to use it.**
36. **Design a non-primitive data for Order with some attributes and show how to use it.**
37. **Design a non-primitive data for Laptop with some attributes and show how to use it.**
38. **Design a non-primitive data for Student with some attributes and show how to use it.**
39. **Design a non-primitive data for Chrome with some attributes and show how to use it.**
40. **Why java is called strictly typed programming language.**

➢ Variable:

41. What is a variable? Why it is used?
42. What are different types of variables?
43. **Explain local, static and non-static variables? How it is accessed in java program?**
44. What is default value of a local variable?
45. What is scope of a local variable?
46. **What is scope of a static variable?**
47. **What is scope of a non-static variable**?
48. Can we have default value for a static or non-static variable?
49. **Can we have a local variable and global (static or non-static) variable with the same name? if so, then which variable gets preference?**
50. **Explain what is variable shadowing?**

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➢ Primitive Type casting:

51. What is primitive type casting?
52. **Explain widening and narrowing?**
53. **Explain implicit and explicit conversion.**
54. Why widening is called implicit conversion and narrowing is called explicit conversion?

Methods:

55. What is method? Why it is used?
56. How many methods can be designed inside a class?
57. How method is executed?
58. In which memory area method execution takes place?
59. Which mechanism is followed in method execution?
60. How we can call and execute a static type method?
61. How we can call and execute a non-static type method?
62. What is the main advantage of a method?
    ➔Code Reusability
63. **Why main method of java should be always declared as public?**
64. **Why main method should be always declared as static?**
65. **What will happen if multiple copy of main method is loaded in memory?**
    //To maintain uniformity in program execution
66. **Why main method return type is always void?**
67. **What is public in method?**
68. **Explain access modifiers?**
69. **How many access modifiers can be used by a method?**
    ➔only one
70. **What is static in method?**
71. **Explain non-access modifiers?**
72. **How many non-access modifiers can be used by a method?**
73. **What is method signature?**
74. **What is method declaration?**
75. **What is method definition?**
76. **What is formal argument and actual argument?**
77. **What is Compile Time Binding/ Early Binding?**
78. **What is the purpose of String[] args in main method?**
    ➔To support Command Line Argument Feature in java

79. If there are multiple methods inside a class along with main method then which method will be executed?
80. If there are multiple methods inside a class without main method then which method will be executed?
81. If there are multiple main methods inside a class with **main(String[] args), main(String args)** and **main(int args)** method then which method will be executed?
82. If there are multiple main methods inside a class with **main(String args), main(int args) and main(int[] args)** method then which method will be executed?
83. What are the types of methods?
84. Which method can be used to transfer values from one method frame to another method frame?
85. What value is returned from a method with return type void?
86. How many values can we return from a method?
87. Which type of method can be directly called inside System.out.println()?
88. **Can we use static and abstract keyword together?**
    ➔No, it forms illegal combination
89. **Explain what is call by value?**
90. **Explain what is call by reference?**
91. **Does java support call by reference?**

➢ JVM, JRE, JDK, JIT Compiler:

92. Explain JVM and its working.
93. Explain JRE.
94. Explain JDK.
95. Explain JIT Compiler.
96. Explain the difference between JVM, JRE and JDK.

➢ Garbage Collection:

97.What is Garbage Collection in Java?

➔Automatic process by JVM to free memory by destroying unwanted members (such as unreachable objects, local variables etc)

98.Why is Garbage Collection needed?

➔Prevents memory leaks and "**OutOfMemoryError"** by reclaiming memory from unused objects.

99.How does Java Garbage Collector identify unused objects?

➔Using Reachability Analysis

100. Can you force garbage collection in Java?

➔You can request it using **System.gc()**, but **it's not guaranteed**. It is just a request to JVM to start garbage collection immediately but JVM can ignore it.

101. Explain the working of garbage collector in java?

➔The Garbage Collector (GC) in Java is an automatic process managed by the Java Virtual Machine (JVM) that reclaims memory by removing objects that are no longer in use.

102. What do you mean by automatic garbage collection in java?

➔ Automatic Garbage Collection in Java means that the Java Virtual Machine (JVM) takes care of memory management by automatically identifying and removing unused objects (garbage) from the heap, without requiring the programmer to explicitly delete them.

103. How Is an Object Deleted in Java?

➔In Java, objects are not explicitly deleted by the programmer like in C/C++ (no delete keyword). Instead, object deletion is handled automatically by the Garbage Collector (GC).

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Once there are **no references** to the object, Object becomes unreachable and it becomes **eligible for garbage collection**.

a)
Student s = new Student(); // Object created in heap memory
S=null; // Now the Student object is unreachable

b)
Student s1 = new Student();
Student s2 = new Student();
s1 = s2; // The original object referenced by s1 is now unreachable and s1 object will be eligible for garbage collection

c)
when stack frame is deleted then all objects of that frame will become unreachable and it will be deleted by garbage collector.

104. What are the advantages of garbage collector?

➔ Advantages of Garbage Collector in Java

1. **Automatic Memory Management**
   - ✓ Developers don't need to explicitly manage memory
2. **Avoids Memory Leaks**
   - ✓ GC cleans unreachable objects automatically.
   - ✓ Prevents common memory leaks caused by forgetting to deallocate memory.
3. **Increased Developer Productivity**
   - ✓ Developers can focus on business logic instead of memory management.
   - ✓ Less boilerplate code → faster development.
4. **Safety and Security**
   - ✓ Makes Java **more secure** and **robust** than manual memory management.
5. **Optimized Memory Usage**

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✓ GC algorithms optimize memory which helps maintain long-running applications with large heaps.

105. What are the disadvantages of garbage collector?

➔While the Garbage Collector (GC) makes Java memory management easier and safer, it comes with some disadvantages as well:

1. Unpredictable Performance (GC Pauses)

- The GC may stop application threads to reclaim memory.
- These pauses can **affect performance**, especially in **real-time** or **latency-sensitive** applications.

Example: In a high-frequency trading app, even a small pause can cause loss of money.

2. Less Control Over Memory

- In languages like C/C++, developers manually allocate and deallocate memory.
- In Java, the GC decides **when** and **what** to clean — developers **can't force** it to run.

Calling **System.gc()** is just a suggestion; the JVM may ignore it.

3. Overhead and CPU Usage

- GC consumes **CPU cycles** to monitor, trace, and clean memory.
- In large applications, this can result in **increased CPU load**.

4. Memory Overhead

- Some GC algorithms may use **extra memory** to manage regions, generations, or threads.
- More **metadata and thread stacks** → more memory consumed.

➢ OOP Programming:

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106. What is Object Oriented programming? Why it is used?
107. What are the four pillars of OOP?
108. Is java a 100% object-oriented language? Explain.
109. What is class and object? Define it.
110. What are the properties of a class?
111. Explain new keyword.
112. Explain how object is created in java?
113. What is the work of constructor?
114. What are reference data type? What is its default value?
115. Explain reference variable? What is its default value?
116. What do you mean by instantiation?
117. What is instance?
118. Explain static type members?
119. What is the main purpose of static members?
➔memory management because only one copy is loaded inside memory for static type members
120. How many copies of static members are loaded in a program execution?
121. Why main method is always static?
122. Explain non-static members?
123. What is the main purpose of non-static members?
➔To load different copy for different object
124. How many copies of non-static members are loaded in a program execution?
125. What are class members?
126. What are instance members?
127. Why static members are called class members?
128. Why non-static members are called instance members?
129. What is the different memory block are created by JVM at run time. Explain them.
130. Explain the difference between Stack and Heap Area?
131. Explain static and non-static block or what is static or non-static initializer block?
132. Can we run a java program without main method?
133. Which type members can be accessed directly inside a static method?
134. Which type members can be accessed directly inside a non-static method?
135. Can we access a static member directly inside static block?
136. Can we access a non-static member directly inside non-static block?
137. Can we access a static member directly inside non-static block?
138. Can we access a non-static member directly inside a static block?
139. Explain variable shadowing?
140. How variable shadowing is resolved in case of static variables?
141. How variable shadowing is resolved in case of non-static variables?
142. Explain this keyword?
143. What will be inside this keyword? OR, what does this keyword contain?
//It contains the reference of the current object under execution

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Constructor:

158. What is constructor? What is its use?
159. Explain the difference between method and constructor?
OR, how does a constructor differ from a method?
160. What is the return type of a constructor?
161. Can a constructor be static or final?
162. What are the different types of constructors?
163. Explain default constructor with its use?
164. Explain parameterized constructor? What is its use?
165. How a constructor is called?
166. Explain constructor overloading with its use?
167. What is the advantage of constructor overloading?
168. What is disadvantage of constructor overloading?
169. Explain "Boilerplate code".
➔A set of code which is repeatedly being used with little or no change
170. Explain constructor chaining with its use?
171. What is the advantage of constructor chaining?
//It helps to reduce boilerplate code
172. Explain this() statement.
173. How many constructors can be called by one constructor?

174. Can we make this() as the second statement?
175. Can a constructor call a method?
➔Yes
176. Can we use this() call inside a method?
➔NO
177. What are the restrictions on constructors?
178. Explain copy constructor with a program. What is its use?

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